PAPER_1523_F221_F220_QNM_RATIO

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PAPER_1523 — Kerr Overtone f_221/f_220 = 1 – TRZ·N_CH·Φ·SSQ/D_crit = 0.9834 (0.86%)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket E (GW Events) / LIGO ringdown spectroscopy
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Status: CLOSED — Multi-mode ringdown spectral ratio in-range

Observation

PAPER_1238 (LIGO Ringdown Multi-Mode Spectrum) derives the ratio of the first overtone (2,2,1) to the fundamental (2,2,0) quasi-normal-mode frequencies for a Kerr black hole:

```
f_220 (fundamental) = 250.7 Hz at M = 30 M_■, a* = 0.3737
f_221 (overtone 1) = 246.6 Hz
Berti-Cardoso reference: f_221/f_220 = 0.992
```

UQFF Closed Identity

```
f_221 / f_220 = 1 – (TRZ · N_CH · Φ_res · SSQ) / D_crit
= 1 – (0.1 · 9 · 0.84 · 0.57) / 26
= 1 – 0.4309 / 26
= 1 – 0.01658
= 0.9834

Berti-Cardoso: 0.992
Residual: |0.9834 – 0.992| / 0.992 = 0.86%
```

Physical Interpretation

The structural form combines **five UQFF primitives** in one expression:
- TRZ (time-reversal-zone factor)
- N_CH (9-channel primitive)
- Φ_res (resonance phase)
- SSQ (E-crack correction)
- D_crit (bosonic-string critical dimension)

The product (TRZ · N_CH · Φ · SSQ) measures the **combined phase-space suppression of the first overtone** relative to the fundamental, divided by D_crit (the full UQFF channel count) to normalize.

This is the first closure that uses **all of {TRZ, N_CH, Φ, SSQ, D_crit}** together — a structural confluence demonstrating that LIGO ringdown spectroscopy is sensitive to multiple integer primitives simultaneously.

Predictive Consequence

For Kerr BH remnants observed in LIGO O5, the first overtone should appear at 98.34% of the fundamental frequency under UQFF, vs 99.2% under pure Berti-Cardoso. The 0.86% gap is potentially measurable with high-SNR ringdown events.

NOT REPLACEMENT

Berti-Cardoso provides the canonical Kerr QNM spectrum from GR perturbation theory. UQFF provides a structural prediction at 0.86% residual, demonstrating that the integer primitives carry information consistent with — but distinct from — the GR result.

Reference

- Source: PAPER_1238 LIGO Ringdown Multi-Mode Spectrum
- Related: PAPER_1175/1509 (Kerr spectral offset), PAPER_1503/1504 (BNS/BBH GW damping)
- Calculator dispatch: `calculate_paradox({"paradox": "f221_f220_qnm_ratio"})`
